



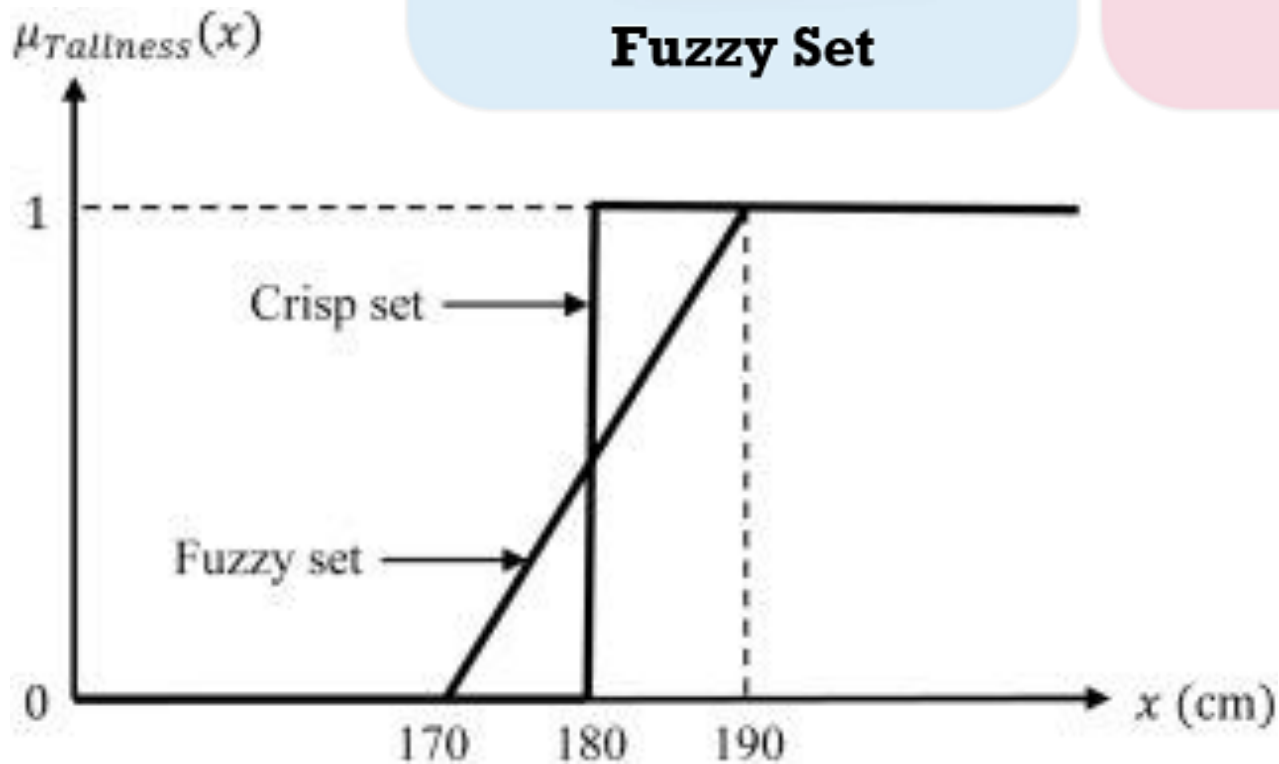
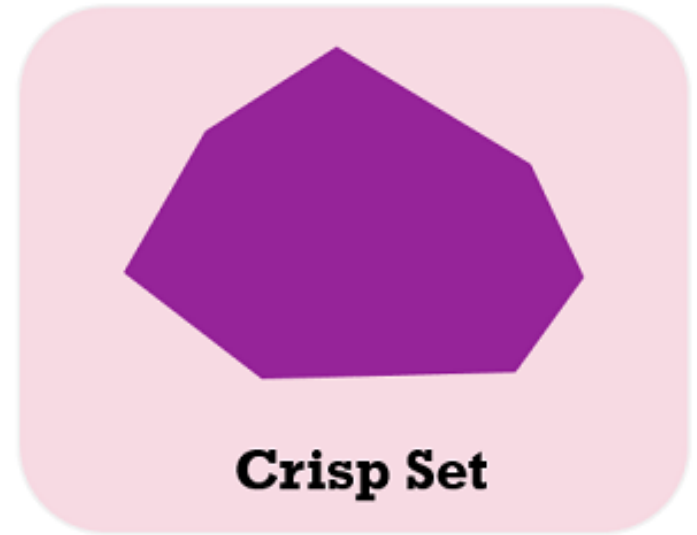
Fuzzy Logic & Neural Networks (CS-514)

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Fuzzy Sets



Fuzzy Sets

universe of discourse



The fuzzy set A in UoD X is defined as a set of ordered pairs:

$$A = \{(x, \mu_A(x)) \mid x \in X\}$$

where $\mu_A(x)$ is membership degree and $\mu_A(x) \in [0, 1]$

Example: The fuzzy set of expensive cars can be described as:

$$A = \{(\text{Ferrari}, 1), (\text{Rolls Royce}, 1), (\text{Mercedes}, 0.8), (\text{BMW}, 0.7), (\text{Honda}, 0.4)\}.$$

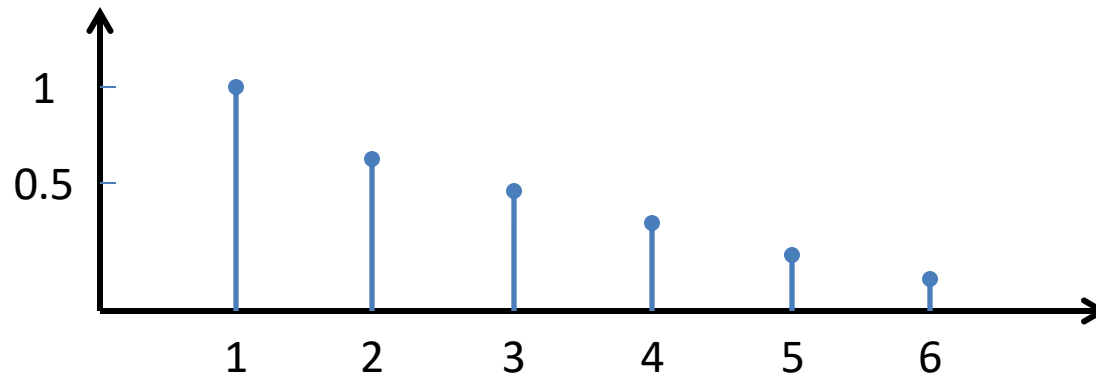
Fuzzy Sets

On Discrete UoD:

The UoD can be discrete in many practical problems. For such cases, a fuzzy set can be represented as:

$$A = 1/1 + 0.8/2 + 0.5/3 + 0.4/4 + 0.3/5 + 0.2/6$$

$$A = \sum_{x_i \in X} \mu_A(x_i) / x_i$$

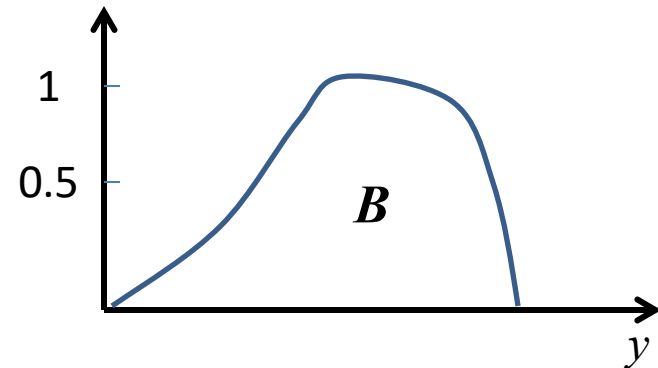
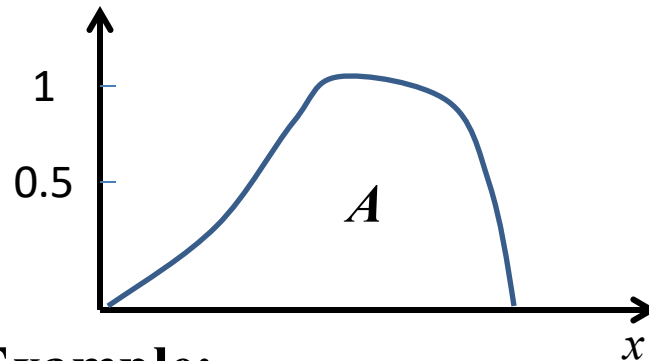


Fuzzy Sets

Operations on Fuzzy Sets:

- **Equality:** Two fuzzy sets A and B can be considered equal if -

$$\mu_A(x) = \mu_B(y), \forall x \in X, y \in Y \text{ and } x=y$$



Example:

$$A(x) = \{(x1,0.1), (x2,0.2), (x3,0.3), (x4,0.4)\}$$

$$B(x) = \{(x1,0.1), (x2,0.5), (x3,0.3), (x4,0.6)\}$$

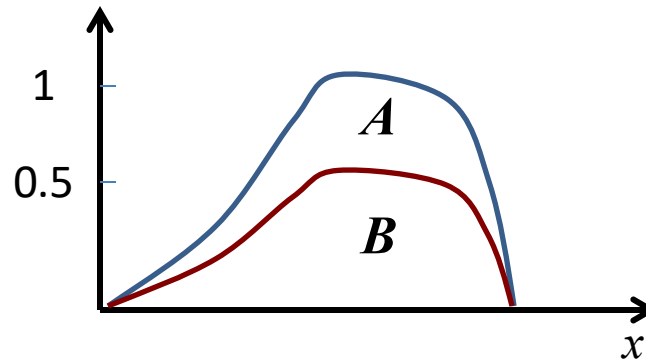
As $\mu_A(x) \neq \mu_B(x)$ for different $x \in X$, $A(x) \neq B(x)$

Fuzzy Sets

Operations on Fuzzy Sets:

- **Containment (Subset):** A fuzzy set B can be considered as a subset of fuzzy set A if -

$$\mu_A(x) \geq \mu_B(y), \forall x \in X, y \in Y \text{ and } x=y$$



Example:

$$A(x) = \{(x1,0.1), (x2,0.2), (x3,0.3), (x4,0.4)\}$$

$$B(x) = \{(x1,0.1), (x2,0.1), (x3,0.2), (x4,0.4)\}$$

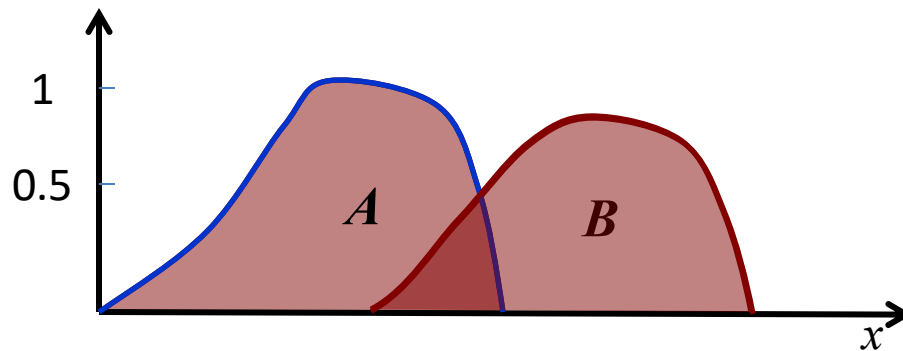
As $\mu_A(x) \geq \mu_B(x)$ for different $x \in X$, $B(x)$ is subset of $A(x)$

Fuzzy Sets

Operations on Fuzzy Sets:

Union : Union of two fuzzy sets A and B can be defined as-

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x)), \quad \forall x \in X$$



Fuzzy Sets

Operations on Fuzzy Sets:

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x)), \quad \forall x \in X$$

Example:

$$A(x) = \{(x1, 0.7), (x2, 0.3), (x3, 0.9), (x4, 0.1)\}$$

$$B(x) = \{(x1, 0.2), (x2, 0.5), (x3, 0.7), (x4, 0.4)\}$$

$$\mu(A \cup B)(x1) = \max\{\mu A(x1), \mu B(x1)\} = \max\{0.7, 0.2\} = 0.7$$

$$\mu(A \cup B)(x2) = \max\{\mu A(x2), \mu B(x2)\} = \max\{0.3, 0.5\} = 0.5$$

$$\mu(A \cup B)(x3) = \max\{\mu A(x3), \mu B(x3)\} = \max\{0.9, 0.7\} = 0.9$$

$$\mu(A \cup B)(x4) = \max\{\mu A(x4), \mu B(x4)\} = \max\{0.1, 0.4\} = 0.4$$

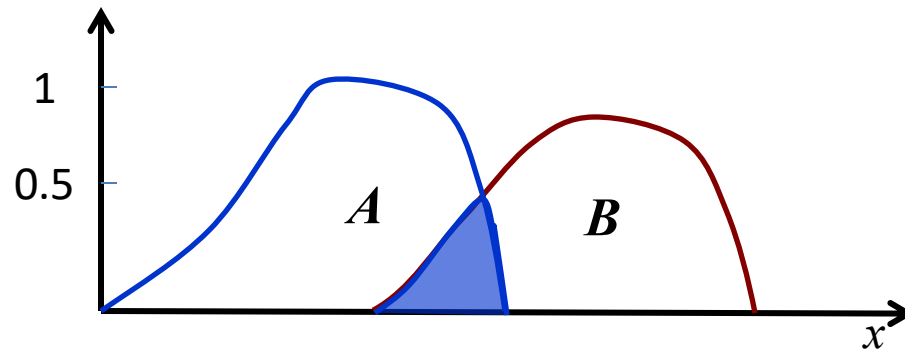
$$A \cup B(x) = \{(x1, 0.7), (x2, 0.5), (x3, 0.9), (x4, 0.4)\}$$

Fuzzy Sets

Operations on Fuzzy Sets:

- **Intersection:** Intersection of two fuzzy sets A and B can be defined as-

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x)), \quad \forall x \in X$$



Fuzzy Sets

Operations on Fuzzy Sets:

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x)), \quad \forall x \in X$$

Example:

$$A(x) = \{(x1, 0.7), (x2, 0.3), (x3, 0.9), (x4, 0.1)\}$$

$$B(x) = \{(x1, 0.2), (x2, 0.5), (x3, 0.7), (x4, 0.4)\}$$

$$\mu(A \cap B)(x1) = \min\{\mu_A(x1), \mu_B(x1)\} = \min\{0.7, 0.2\} = 0.2$$

$$\mu(A \cap B)(x2) = \min\{\mu_A(x2), \mu_B(x2)\} = \min\{0.3, 0.5\} = 0.3$$

$$\mu(A \cap B)(x3) = \min\{\mu_A(x3), \mu_B(x3)\} = \min\{0.9, 0.7\} = 0.7$$

$$\mu(A \cap B)(x4) = \min\{\mu_A(x4), \mu_B(x4)\} = \min\{0.1, 0.4\} = 0.1$$

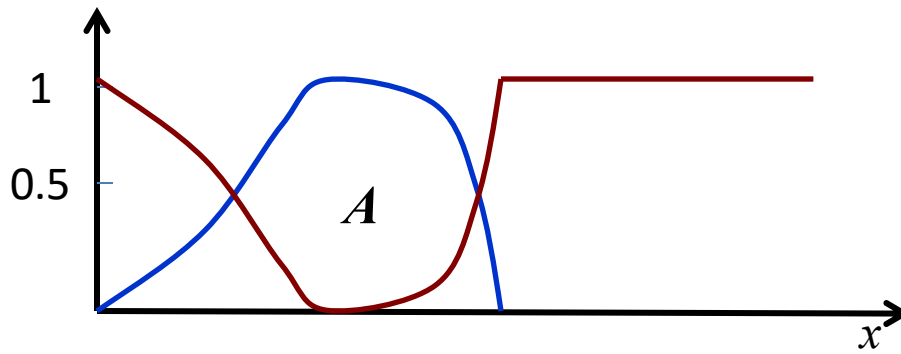
$$A \cap B(x) = \{(x1, 0.2), (x2, 0.3), (x3, 0.7), (x4, 0.1)\}$$

Fuzzy Sets

Operations on Fuzzy Sets:

- **Compliment:** Fuzzy complement of a fuzzy set A can be defined as-

$$\overline{\mu}_A(x) = (1 - \mu_A(x)), \quad \forall x \in X$$



Fuzzy Sets

Governing Rules:

Commutativity $A \cup B = B \cup A$

$$A \cap B = B \cap A$$

Associativity $A \cup (B \cup C) = (A \cup B) \cup C$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributivity $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Idempotency $A \cup A = A$

$$A \cap A = A$$

Identity $A \cup \emptyset = A$

$$A \cap X = A$$

$$A \cap \emptyset = \emptyset$$

$$A \cup X = X$$

Absorption of compliments

$$A \cup (\bar{A} \cap C) = (A \cup C)$$

$$A \cap (\bar{A} \cup C) = (A \cap C)$$

Fuzzy Sets

Governing Rules:

Transitivity If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$

Involution

$$\bar{\bar{A}} = A$$

Law of Excluded Middle

$$A \cup \bar{A} = X$$

Law of Contradiction

$$A \cap \bar{A} = \emptyset$$

DeMorgan's Laws

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B}$$

$$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

Assignment

With suitable graphical examples prove following properties on fuzzy sets

Law of Excluded Middle

$$A \cup \bar{A} = X$$

De Morgan's Laws

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B}$$

Fuzzy Sets

Operations on Fuzzy Sets:

- **Algebraic product of fuzzy sets:** The Algebraic product of two fuzzy sets $A(x)$ and $B(x)$ for all $x \in X$, is denoted by $A(x).B(x)$ and defined as follows:

$$A(x).B(x) = \{(x, \mu_A(x).\mu_B(x)), x \in X\}$$

Example:

Fuzzy Sets

Operations on Fuzzy Sets:

- **Multiplication of fuzzy sets by a crisp number:** The product of fuzzy set $A(x)$ and a crisp number 'd' is expressed as follows:

$$A(x).B(x) = \{(x, d \cdot \mu_A(x)), x \in X\}$$

Example:

Fuzzy Sets

Operations on Fuzzy Sets:

- **Power of a fuzzy set:** The p^{th} power of a fuzzy set $A(x)$ yields another fuzzy set $A^P(x)$, whose value can be determined as follows:

$$A^P(x) = \{(x, (\mu_A(x))^P), x \in X\}$$

Example:

Fuzzy Sets

Operations on Fuzzy Sets:

- **Algebraic sum of two fuzzy sets:** The Algebraic sum of two fuzzy sets $A(x)$ and $B(x)$ for all $x \in X$, is denoted by $A(x)+B(x)$ and defined as follows:

$$A(x)+B(x)= \{(x, \mu_{A+B}(x)), x \in X \}$$

where $\mu_{A+B}(x) = \mu_A(x)+\mu_B(x) - \mu_A(x).\mu_B(x)$

Example:

$$A(x)=\{(x1,0.1),(x2,0.2),(x3,0.3),(x4,0.4)\}$$

$$B(x)=\{(x1,0.5),(x2,0.7),(x3,0.8),(x4,0.9)\}$$

$$A(x)+B(x) = \{(x1,0.55), (x2,0.76), (x3,0.86), (x4,0.94)\}$$

Fuzzy Sets

Operations on Fuzzy Sets:

- **Bounded sum of two fuzzy sets:** The bounded sum of two fuzzy sets $A(x)$ and $B(x)$ for all $x \in X$, is denoted by

$A(x) \oplus B(x)$ and defined as follows:

$$A(x) \oplus B(x) = \{(x, \mu_{A \oplus B}(x)), x \in X\}$$

where $\mu_{A \oplus B}(x) = \min\{1, \mu_A(x) + \mu_B(x)\}$

Example:

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.4)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8), (x_4, 0.9)\}$$

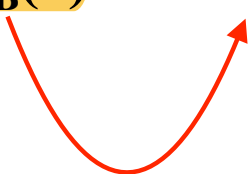
Fuzzy Sets

Operations on Fuzzy Sets:

- **Algebraic deference of two fuzzy sets:** The Algebraic deference of two fuzzy sets $A(x)$ and $B(x)$ for all $x \in X$, is denoted by $A(x)-B(x)$ and defined as follows:

$$A(x) - B(x) = \{(x, \mu_{A-B}(x), x \in X \}$$

where $\mu_{A-B}(x) = \mu_{A \cap \bar{B}}(x)$ $B(\text{complement}) = 1-B$



Example:

$$A(x) = \{(x1, 0.1), (x2, 0.2), (x3, 0.3), (x4, 0.4)\}$$

$$B(x) = \{(x1, 0.5), (x2, 0.7), (x3, 0.8), (x4, 0.9)\}$$

$$A(x) - B(x) = \{(x1, 0.1), (x2, 0.2), (x3, 0.2), (x4, 0.1)\}$$

Fuzzy Sets

Operations on Fuzzy Sets:

- **Bounded difference of two fuzzy sets:** The bounded difference of two fuzzy sets $A(x)$ and $B(x)$ for all $x \in X$, is denoted by $A(x) \ominus B(x)$ and defined as follows:

$$A(x) \ominus B(x) = \{(x, (\mu_{A \ominus B}(x))), x \in X\}$$

where $\mu_{A \ominus B}(x) = \max\{0, \mu_A(x) + \mu_B(x) - 1\}$

Example:

$$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.4)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8), (x_4, 0.9)\}$$

$$A(x) \ominus B(x) = \{(x_1, 0), (x_2, 0), (x_3, 0.1), (x_4, 0.3)\}$$

Fuzzy Sets

Operations on Fuzzy Sets:

- **Cartesian product of two fuzzy sets:** Let us consider two fuzzy sets $A(x)$ and $B(y)$ defined on the Universal sets X and Y , respectively. The Cartesian product of fuzzy sets $A(x)$ and $B(y)$, is denoted by $A(x) \times B(y)$, such that $x \in X$, $y \in Y$. It is determined, so that the following conditions satisfy:

$$\mu_{A \times B}(x, y) = \min\{\mu_A(x), \mu_B(y)\}$$

Example:

$$A(x) = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5), (x_4, 0.6)\}$$

$$B(y) = \{(y_1, 0.8), (y_2, 0.6), (y_3, 0.3)\}$$

Fuzzy Relation

- A Crisp relation represents presence or absence of association, interaction or interconnection between elements of 2 or more sets.
- This concept can be generalized to various degrees or strengths of association or interaction between elements.
- A fuzzy relation generalizes these degrees to membership grades. So, a crisp relation is a restricted case of a fuzzy relation.

Fuzzy Relation

Crisp relation

Let the Cartesian product between two crisp sets A and B is denoted as $A \times B$, which is a collection of order pairs, such that:

$$A \times B = \{(x, y), (x \in A \text{ and } y \in B)\}$$

Fuzzy Relation

Crisp relation

Example: Consider the two crisp sets A and B as given below.

$$A = \{1, 2, 3, 4\} \quad B = \{3, 5, 7\}$$

Then,

$$A \times B = \{ (1, 3) , (1, 5) , (1, 7) , (2, 3) , (2, 5) , (2, 7) , (3, 3) , (3, 5) , (3, 7) , (4, 3) , (4, 5) , (4, 7) \}$$

Let us define a relation as:

$$R = \{(x, y) \mid y = x + 1, (x, y) \in A \times B\}$$

Therefore,

$$R = \{ (2, 3) , (4, 5) \}$$

Fuzzy Relation

Crisp relation

For the same relation:

$$R = \{(x, y) \mid y = x + 1, (x, y) \in A \times B\}$$

The $R = \{ (2, 3) , (4, 5) \}$, in a matrix form as,

$$R = \begin{matrix} & \begin{matrix} 3 & 5 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Fuzzy Relation

- A Fuzzy relation is a fuzzy set defined on the Cartesian product of sets $X_1, X_2, \dots, X_n$.
- Here, n-tuples $(x_1, x_2, \dots, x_n)$ may have varying degree of memberships within the relationship.
- The membership values indicate the strength of the relation between the tuples.

Fuzzy Relation

Fuzzy relation

Example:

$$X = \{typhoid, viral, cold\}$$

$$Y = \{running\ nose, high\ temp, shivering\}$$

		<i>running nose</i>	<i>high temperature</i>	<i>shivering</i>
$R =$	<i>typhoid</i>	0.1	0.9	0.8
	<i>viral</i>	0.2	0.9	0.7
	<i>cold</i>	0.9	0.4	0.6

Fuzzy Relation

Operations on Fuzzy Relations

Let R and S be two fuzzy relations on $A \times B$.

- Union: $\mu_{R \cup S}(x, y) = \max\{\mu_R(x, y), \mu_S(x, y)\}$
- Intersection: $\mu_{R \cap S}(x, y) = \min\{\mu_R(x, y), \mu_S(x, y)\}$
- Complement: $\mu_{\underline{R}}(x, y) = 1 - \mu_R(x, y)$

Fuzzy Relation

Operations on Fuzzy Relations

If R and S be two fuzzy relations then:

- Composition:

$$T = R \circ S$$

$$\mu_{R \circ S}(x, z) = \max_{y \in Y} \{ \min (\mu_R(x, y) , \mu_S(y, z)) \}$$

It is also known as **max-min** composition

Fuzzy Relation

Fuzzy Composition (max-min)

Example:

$$X = (x_1, x_2, x_3),$$

$$Y = (y_1, y_2),$$

$$Z = (z_1, z_2, z_3),$$

$$R = \begin{matrix} & y_1 & y_2 \\ x_1 & \begin{bmatrix} 0.5 & 0.1 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.2 & 0.9 \end{bmatrix} \\ x_3 & \begin{bmatrix} 0.8 & 0.6 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & z_1 & z_2 & z_3 \\ y_1 & \begin{bmatrix} 0.6 & 0.4 & 0.7 \end{bmatrix} \\ y_2 & \begin{bmatrix} 0.5 & 0.8 & 0.9 \end{bmatrix} \end{matrix}$$

$$\text{Find } T = R \circ S$$

Fuzzy Relation

Fuzzy Composition

Example:

$$R = \begin{matrix} & y_1 & y_2 \\ x_1 & \begin{bmatrix} 0.5 & 0.1 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.2 & 0.9 \end{bmatrix} \\ x_3 & \begin{bmatrix} 0.8 & 0.6 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & z_1 & z_2 & z_3 \\ y_1 & \begin{bmatrix} 0.6 & 0.4 & 0.7 \end{bmatrix} \\ y_2 & \begin{bmatrix} 0.5 & 0.8 & 0.9 \end{bmatrix} \end{matrix}$$

Find $T = R \circ S$

$$\mu_{R \circ S}(x_1, z_1) = \max \{ \min(\mu_R(x_1, y_1), \mu_S(y_1, z_1)), \min(\mu_R(x_1, y_2), \mu_S(y_2, z_1)) \}$$

$$= \max \{ \min(0.5, 0.6), \min(0.1, 0.5) \}$$

$$= \max \{ 0.5, 0.1 \} = 0.5$$

Fuzzy Relation

Fuzzy Composition

Example:

$$R = \begin{matrix} & y_1 & y_2 \\ x_1 & \begin{bmatrix} 0.5 & 0.1 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.2 & 0.9 \end{bmatrix} \\ x_3 & \begin{bmatrix} 0.8 & 0.6 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & z_1 & z_2 & z_3 \\ y_1 & \begin{bmatrix} 0.6 & 0.4 & 0.7 \end{bmatrix} \\ y_2 & \begin{bmatrix} 0.5 & 0.8 & 0.9 \end{bmatrix} \end{matrix}$$

Find $T = R \circ S$

$$R \circ S = \begin{matrix} & z_1 & z_2 & z_3 \\ x_1 & \begin{bmatrix} 0.5 & 0.4 & 0.5 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.5 & 0.8 & 0.9 \end{bmatrix} \\ x_3 & \begin{bmatrix} 0.6 & 0.6 & 0.7 \end{bmatrix} \end{matrix}$$

Fuzzy Relation

Operations on Fuzzy Relations

If R and S be two fuzzy relations, then:

- Composition:

$$T = R \circ S$$

$$\mu_{R \circ S}(x, z) = \max_{y \in Y} \{ (\mu_R(x, y) \cdot \mu_S(y, z)) \}$$

It is also known as **max-dot** or **max-product** composition

Fuzzy Relation

Fuzzy Composition (max-product)

Example:

Set $A=\{a_1,a_2\}$, Set $B=\{b_1,b_2\}$, Set $C=\{c_1,c_2\}$

Fuzzy Relation R_{AB} between A and B :

$$R_{AB} = \begin{bmatrix} 0.4 & 0.6 \\ 0.7 & 0.2 \end{bmatrix}$$

This means, for example, $R_{AB}(a_1, b_1) = 0.4$, $R_{AB}(a_1, b_2) = 0.6$, and so on.

Fuzzy Relation R_{BC} between B and C :

$$R_{BC} = \begin{bmatrix} 0.5 & 0.8 \\ 0.3 & 0.7 \end{bmatrix}$$

This means, for example, $R_{BC}(b_1, c_1) = 0.5$, $R_{BC}(b_1, c_2) = 0.8$, and so on.

Fuzzy Relation

Max-Product Composition R_{AC} :

To compute the max-product composition R_{AC} , we calculate for each pair (a_i, c_j) :

1. For $R_{AC}(a_1, c_1)$:

$$R_{AC}(a_1, c_1) = \max(0.4 \cdot 0.5, 0.6 \cdot 0.3) = \max(0.2, 0.18) = 0.2$$

2. For $R_{AC}(a_1, c_2)$:

$$R_{AC}(a_1, c_2) = \max(0.4 \cdot 0.8, 0.6 \cdot 0.7) = \max(0.32, 0.42) = 0.42$$

3. For $R_{AC}(a_2, c_1)$:

$$R_{AC}(a_2, c_1) = \max(0.7 \cdot 0.5, 0.2 \cdot 0.3) = \max(0.35, 0.06) = 0.35$$

4. For $R_{AC}(a_2, c_2)$:

$$R_{AC}(a_2, c_2) = \max(0.7 \cdot 0.8, 0.2 \cdot 0.7) = \max(0.56, 0.14) = 0.56$$

Fuzzy Relation

Fuzzy Composition (max-product)

Example:

Set $A=\{a_1,a_2\}$, Set $B=\{b_1,b_2\}$, Set $C=\{c_1,c_2\}$

$$R_{AB} = \begin{bmatrix} 0.4 & 0.6 \\ 0.7 & 0.2 \end{bmatrix}$$

$$R_{BC} = \begin{bmatrix} 0.5 & 0.8 \\ 0.3 & 0.7 \end{bmatrix}$$

Final Composition Matrix R_{AC} :

$$R_{AC} = \begin{bmatrix} 0.2 & 0.42 \\ 0.35 & 0.56 \end{bmatrix}$$

This is the resulting fuzzy relation between sets A and C using the max-product composition method.

FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

A fuzzy relation, R on a single UoD X is also a relation from X to X . It is a fuzzy equivalence relation if all three of the following properties for matrix relations define it:

$$\text{Reflexivity } \mu_R(x_i, x_i) = 1$$

$$\text{Symmetry } \mu_R(x_i, x_j) = \mu_R(x_j, x_i)$$

$$\text{Transitivity } \mu_R(x_i, x_j) = \lambda_1 \text{ and } \mu_R(x_j, x_k) = \lambda_2$$

$$\rightarrow \mu_R(x_i, x_k) = \lambda. \quad \text{where } \lambda \geq \min[\lambda_1, \lambda_2].$$

FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

In fuzzy logic, a tolerance relation satisfies **reflexivity** and **symmetry**, but not necessarily **transitivity**.

Reflexivity $\mu_R(x_i, x_i) = 1$

Symmetry $\mu_R(x_i, x_j) = \mu_R(x_j, x_i)$

Transitivity $\mu_R(x_i, x_j) = \lambda_1$ and $\mu_R(x_j, x_k) = \lambda_2$

$\rightarrow \mu_R(x_i, x_k) = \lambda.$ where $\lambda < \min[\lambda_1, \lambda_2].$

FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

Example:

Consider a fuzzy relation R on a set of numbers $\{1,2,3,4\}$ representing some "closeness" between numbers, with values in the range $[0,1]$. The fuzzy relation $R(x_i, x_j)$ shows how "similar" x_i and x_j are.

$$R = \begin{pmatrix} 1 & 0.8 & 0.6 & 0.2 \\ 0.8 & 1 & 0.7 & 0.3 \\ 0.6 & 0.7 & 1 & 0.5 \\ 0.2 & 0.3 & 0.5 & 1 \end{pmatrix}$$

FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

Solution:

$$R = \begin{pmatrix} 1 & 0.8 & 0.6 & 0.2 \\ 0.8 & 1 & 0.7 & 0.3 \\ 0.6 & 0.7 & 1 & 0.5 \\ 0.2 & 0.3 & 0.5 & 1 \end{pmatrix}$$

Compute Reflexivity $\mu_R(x_i, x_i)$

Symmetry $\mu_R(x_i, x_j) = \mu_R(x_j, x_i)$

Transitivity $\mu_R(x_i, x_j) = \lambda_1$ and $\mu_R(x_j, x_k) = \lambda_2$

$\rightarrow \mu_R(x_i, x_k) = \lambda.$ where $\lambda \geq \min[\lambda_1, \lambda_2].$

FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

Example:

Consider a fuzzy relation R. Find if it satisfy the Fuzzy Tolerance or Equivalence Relations

$$R = \begin{bmatrix} 1 & 0.8 & 0.4 & 0.5 & 0.8 \\ 0.8 & 1 & 0.4 & 0.5 & 0.9 \\ 0.4 & 0.4 & 1 & 0.4 & 0.4 \\ 0.5 & 0.5 & 0.4 & 1 & 0.5 \\ 0.8 & 0.9 & 0.4 & 0.5 & 1 \end{bmatrix}$$

FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

Example: Let the set of temperatures be:
 $\{20^{\circ}\text{C}, 21^{\circ}\text{C}, 23^{\circ}\text{C}, 25^{\circ}\text{C}\}.$

- We want to represent how "similar" or "equivalent" different temperatures are to each other using fuzzy relations.
- The degree of similarity (or equivalence) between temperatures will range from 0 to 1, where:
- **1** means the temperatures are exactly the same.
- **0** means the temperatures are completely dissimilar.

FUZZY TOLERANCE AND EQUIVALENCE RELATIONS

Example: Let the set of temperatures be:
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Matrix for Fuzzy Tolerance Relation R :

$$R = \begin{pmatrix} 1 & 0.9 & 0.6 & 0.2 \\ 0.9 & 1 & 0.7 & 0.3 \\ 0.6 & 0.7 & 1 & 0.5 \\ 0.2 & 0.3 & 0.5 & 1 \end{pmatrix}$$